

LAB 4- Introduction to Kubernetes on Google Cloud using GKE

Lab Task 1 – Basic Kubernetes operations

After creating the google account, installing the google sdk and kubectl, I created a node -1 cluster- onlineboutique

```
oadewusi@oluSec:~/aiops/Assignments$ gcloud container clusters create onlineboutique \
--project=aiops4 \
--zone=europe-west4-a \
--machine-type=e2-standard-4 \
--num-nodes=1 \
--workload-pool aiops4.svc.id.goog \
--gateway-api "standard"
Note: The Kubelet readonly port (10255) is now deprecated. Please update your workloads to use the recommended alternatives. See https://cloud.google.com/kubernetes-engine/docs/how-to/disable-kubelet-readonly-port for ways to check usage and for migration instructions.
Note: Your Pod address range (--cluster-ipv4-cidr) can accommodate at most 1008 node(s).
Creating cluster onlineboutique in europe-west4-a... Cluster is being health-checked (Kubernetes Control Plane is healthy)...done.
Created [https://container.googleapis.com/v1/projects/aiops4/zones/europe-west4-a/clusters/onlineboutique].
To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workload/_gcloud/europe-west4-a/onlineboutique?project=aiops4
kubeconfig entry generated for onlineboutique.
NAME LOCATION MASTER_VERSION MASTER_IP MACHINE_TYPE NODE_VERSION NUM_NODES STATUS
onlineboutique europe-west4-a 1.30.3-gke.1969001 34.34.95.197 e2-standard-4 1.30.3-gke.1969001 1 RUNNING
```

Also downloaded and unzipping the bookinfo file from GitHub before applying the yaml file

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl apply -f platform/kube/bookinfo.yaml
service/details created
serviceaccount/bookinfo-details created
deployment.apps/details-v1 created
service/ratings created
serviceaccount/bookinfo-ratings created
deployment.apps/ratings-v1 created
service/reviews created
serviceaccount/bookinfo-reviews created
deployment.apps/reviews-v1 created
deployment.apps/reviews-v2 created
deployment.apps/reviews-v3 created
service/productpage created
serviceaccount/bookinfo-productpage created
deployment.apps/productpage-v1 created
```

kubectl get services:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get services
NAME         TYPE        CLUSTER-IP      EXTERNAL-IP      PORT(S)        AGE
details      ClusterIP   34.118.229.26   <none>           9080/TCP       3m5s
kubernetes   ClusterIP   34.118.224.1    <none>           443/TCP        13m
productpage   ClusterIP   34.118.232.94   <none>           9080/TCP       3m
ratings       ClusterIP   34.118.235.11   <none>           9080/TCP       3m4s
reviews       ClusterIP   34.118.238.209  <none>           9080/TCP       3m2s
```

Changed the productpage to loadbalancer with : **kubectl patch svc productpage -p '{"spec": {"type": "LoadBalancer"}}'**

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl patch svc productpage -p '{"spec": {"type": "LoadBalancer"}}'
service/productpage patched
```

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get services
NAME         TYPE        CLUSTER-IP      EXTERNAL-IP      PORT(S)        AGE
details      ClusterIP   34.118.229.26   <none>           9080/TCP       6m1s
kubernetes   ClusterIP   34.118.224.1    <none>           443/TCP        16m
productpage   LoadBalancer 34.118.232.94   34.90.17.208    9080:32237/TCP 5m56s
ratings       ClusterIP   34.118.235.11   <none>           9080/TCP       6m
reviews       ClusterIP   34.118.238.209  <none>           9080/TCP       5m58s
```

Accessed it via my web browser with its public IP:

BookInfo Sample

The Comedy of Errors

Wikipedia Summary: The Comedy of Errors is one of **William Shakespeare's** early plays. It is his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.

[Learn more about Istio →](#)

Book Details

ISBN-10	Publisher	Pages	Type	Language
1234567890	PublisherA	200	paperback	English

Book Reviews

★★★★★ ★★★★★☆

Kubernetes Services: A service is a concept that specifies a policy for accessing Pods and an orderly group of Pods. Flexible interaction between dependent Pods is made possible by services. They give customers consistent endpoints to communicate with, regardless of the creation or destruction of Pods.

Key Functions:

Load Balancing: Distributes traffic across multiple Pods.

Service Discovery: Provides a consistent way for clients to connect to Pods.

Types of Services:

ClusterIP: Exposes the service on an internal IP in the cluster.

NodePort: Exposes the service on each Node's IP at a static port.

LoadBalancer: Provisions an external load balancer (as seen in your setup).

Commands:

List services: `kubectl get services`

Describe a service: `kubectl describe service <service-name>`

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get services
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
details       ClusterIP      34.118.229.26    <none>           9080/TCP         3m5s
kubernetes    ClusterIP      34.118.224.1     <none>           443/TCP          13m
productpage   ClusterIP      34.118.232.94    <none>           9080/TCP         3m
ratings       ClusterIP      34.118.235.11    <none>           9080/TCP         3m4s
reviews       ClusterIP      34.118.238.209   <none>           9080/TCP         3m2s
```

```

● oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl describe svc productpage
Name:                productpage
Namespace:            default
Labels:               app=productpage
                     service=productpage
Annotations:          cloud.google.com/neg: {"ingress":true}
Selector:             app=productpage
Type:                 LoadBalancer
IP Family Policy:     SingleStack
IP Families:          IPv4
IP:                   34.118.232.94
IPs:                  34.118.232.94
LoadBalancer Ingress: 34.90.17.208
Port:                 http 9080/TCP
TargetPort:           9080/TCP
NodePort:             http 32237/TCP
Endpoints:            10.76.0.18:9080
Session Affinity:     None
External Traffic Policy: Cluster
Events:
  Type    Reason          Age          From                    Message
  ----    -
  Normal  ADD             19m          sc-gateway-controller   default/productpage
  Normal  Type            15m          service-controller      ClusterIP -> LoadBalancer
  Normal  UPDATE          15m (x2 over 15m)  sc-gateway-controller   default/productpage
  Normal  EnsuringLoadBalancer 15m          service-controller      Ensuring load balancer
  Normal  EnsuredLoadBalancer 15m          service-controller      Ensured load balancer

```

Deployments: A Deployment manages a set of identical Pods, ensuring the desired number of replicas are running. It provides declarative updates for Pods and Replica sets.

Key Functions:

Scaling: Automatically adjusts the number of running Pods.

Rolling Updates: Ensures zero downtime by gradually replacing old Pods with new ones.

Self-Healing: Replaces failed Pods automatically.

Commands:

List deployments: `kubectl get deployments`

Describe a deployment: `kubectl describe deployment <deployment-name>`

```

● oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
details-v1          1/1     1             1           26m
productpage-v1      1/1     1             1           26m
ratings-v1          1/1     1             1           26m
reviews-v1          1/1     1             1           26m
reviews-v2          1/1     1             1           26m
reviews-v3          1/1     1             1           26m
○ oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ 

```

```

● oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl describe deployment productpage-v1
Name:                productpage-v1
Namespace:           default
CreationTimestamp:    Wed, 02 Oct 2024 12:04:55 +0200
Labels:              app=productpage
                    version=v1
Annotations:         deployment.kubernetes.io/revision: 1
Selector:             app=productpage,version=v1
Replicas:            1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:        RollingUpdate
MinReadySeconds:      0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:             app=productpage
                    version=v1
  Annotations:        prometheus.io/path: /metrics
                    prometheus.io/port: 9080
                    prometheus.io/scrape: true
  Service Account:    bookinfo-productpage
  Containers:
    productpage:
      Image:           docker.io/istio/examples-bookinfo-productpage-v1:1.20.2
      Port:            9080/TCP
      Host Port:       0/TCP
      Environment:     <none>
      Mounts:
        /tmp from tmp (rw)
  Volumes:
    tmp:
      Type:            EmptyDir (a temporary directory that shares a pod's lifetime)
      Medium:
      SizeLimit:       <unset>
  Node-Selectors:     <none>
  Tolerations:        <none>

```

Pods: A Pod is the smallest deployable unit in Kubernetes. It encapsulates one or more containers, storage resources, a unique network IP, and options that govern how the container(s) should run.

Key Functions:

Isolation: Provides an isolated environment for containers.

Networking: Each Pod gets its own IP address.

Lifecycle Management: Handles starting, stopping, and restarting containers.

Commands:

List pods: `kubectl get pods`

Describe a pod: `kubectl describe pod <pod-name>`

View logs for troubleshooting: `kubectl logs <pod-name>`

```

● oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
details-v1-6cd6d9df6b-dt9tj        1/1     Running   0           29m
productpage-v1-57ffb6658c-x4gzd     1/1     Running   0           29m
ratings-v1-794744f5fd-fhmqb        1/1     Running   0           29m
reviews-v1-67896867f4-j6clg        1/1     Running   0           29m
reviews-v2-86d5db4bd6-2rdsx        1/1     Running   0           29m
reviews-v3-77947c4c78-mhmqh9       1/1     Running   0           29m

```

```

oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl describe pods productpage-v1-57ffb6658c-x4gzd
Name:          productpage-v1-57ffb6658c-x4gzd
Namespace:     default
Priority:       0
Service Account: bookinfo-productpage
Node:          gke-onlineboutique-default-pool-35c278d7-2wfs/10.164.0.3
Start Time:    Wed, 02 Oct 2024 12:04:55 +0200
Labels:        app=productpage
               pod-template-hash=57ffb6658c
               version=v1
Annotations:   prometheus.io/path: /metrics
               prometheus.io/port: 9080
               prometheus.io/scrape: true
Status:        Running
IP:            10.76.0.18
IPs:           10.76.0.18
Controlled By: ReplicaSet/productpage-v1-57ffb6658c
Containers:
  productpage:
    Container ID:  containerd://31eb4fd882166381af91caf1e8fba8a6efd93d311697f3ba056893b792dcd4de
    Image:         docker.io/istio/examples-bookinfo-productpage-v1:1.20.2
    Image ID:      docker.io/istio/examples-bookinfo-productpage-v1@sha256:f3754ad97d9abfef9b92913dd57f3a6b5b1ede74a77b0fd699fa41984fa1d3a4
    Port:          9080/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Wed, 02 Oct 2024 12:05:13 +0200
    Ready:         True
    Restart Count:  0
    Environment:   <none>
    Mounts:
      /tmp from tmp (rw)
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-wtm8r (ro)
Conditions:

```

```

oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl logs productpage-v1-57ffb6658c-x4gzd
[2024-10-02 10:05:13 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2024-10-02 10:05:13 +0000] [1] [INFO] Listening at: http://[::]:9080 (1)
[2024-10-02 10:05:13 +0000] [1] [INFO] Using worker: gevent
[2024-10-02 10:05:13 +0000] [6] [INFO] Booting worker with pid: 6
[2024-10-02 10:05:13 +0000] [7] [INFO] Booting worker with pid: 7
[2024-10-02 10:05:13 +0000] [8] [INFO] Booting worker with pid: 8
[2024-10-02 10:05:14 +0000] [9] [INFO] Booting worker with pid: 9
[2024-10-02 10:05:14 +0000] [10] [INFO] Booting worker with pid: 10
[2024-10-02 10:05:14 +0000] [11] [INFO] Booting worker with pid: 11
[2024-10-02 10:05:14 +0000] [12] [INFO] Booting worker with pid: 12
[2024-10-02 10:05:14 +0000] [13] [INFO] Booting worker with pid: 13

```

Changing the cluster's configuration by scaling up the ratings-v1 deployment to two Replica sets

```

oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get deployment ratings-v1
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
ratings-v1    1/1     1             1           42m

oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl scale deployment ratings-v1 --replicas=2
deployment.apps/ratings-v1 scaled

oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get deployment ratings-v1
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
ratings-v1    2/2     2             2           42m

```

Verification of the deployment has been updated:

```

oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
details-v1-6cd6d9df6b-dt9tj         1/1     Running   0           43m
productpage-v1-57ffb6658c-x4gzd      1/1     Running   0           43m
ratings-v1-794744f5fd-5tgz2         1/1     Running   0           76s
ratings-v1-794744f5fd-fhmqb         1/1     Running   0           43m
reviews-v1-67896867f4-j6clg         1/1     Running   0           43m
reviews-v2-86d5db4bd6-2rdsx         1/1     Running   0           43m
reviews-v3-77947c4c78-mhnh9         1/1     Running   0           43m

```

Issuing of `kubectl logs <new pod ID>` to verify it's up and running and listening on port 9080.

```

oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl logs ratings-v1-794744f5fd-5tgz2
Server listening on: http://0.0.0.0:9080

```

Scale ratings-v1 deployment back down to 1 replica:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl scale deployment ratings-v1 --replicas=1
deployment.apps/ratings-v1 scaled
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
details-v1-6cd6d9df6b-dt9tj         1/1     Running   0           48m
productpage-v1-57ffb6658c-x4gzd      1/1     Running   0           48m
ratings-v1-794744f5fd-fhmqb         1/1     Running   0           48m
reviews-v1-67896867f4-j6clg         1/1     Running   0           48m
reviews-v2-86d5db4bd6-2rdsx         1/1     Running   0           48m
reviews-v3-77947c4c78-mhnh9         1/1     Running   0           48m
```

Lab Task 2 – Build and deploy a new image to GKE

I edited the bookinfo/src/productpage/templates/productpage.html file and inserted new text into the block title as shown:

```
28 </script>
29 {% block title %}My Modified BookInfo App{% endblock %}
30
31 <nav class="bg-gray-800">
32   <div class="container mx-auto px-4 sm:px-6 lg:px-8">
33     <div class="relative flex h-16 items-center justify-between">
34       <a href="#" class="text-white px-3 py-2 text-lg font-medium aria-current="page">BookInfo_Aiops</a>
35       <div class="absolute inset-y-0 right-0 flex items-center pr-2 sm:static sm:inset-auto sm:ml-6 sm:pr-0">
36         {% if user: %}
37         <a href="#" class="group block flex-shrink-0">
38           <div class="flex items-center">
```

Defined two environment variables:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ export BOOKINFO_HUB=oadewusi/productpage
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ export BOOKINFO_TAG=latest
```

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src$ ./build-services.sh
++ dirname ./build-services.sh
++ cd .
++ pwd
+ SCRIPTDIR=/home/oadewusi/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src
+ cd /home/oadewusi/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src/../../..
+ h=oadewusi/productpage
+ t=latest
+ [[ oadewusi/productpage == \i\s\t\i\o ]]
+ [[ oadewusi/productpage == \d\o\c\k\k\e\r\.\i\o\.\i\s\t\i\o ]]
+ [[ ' ' == \t\r\u\e ]]
+ env TAGS=latest HUB=oadewusi/productpage docker buildx bake -f samples/bookinfo/src/docker-bake.hcl
[+] Building 28.7s (1/2)
=> [internal] load local bake definitions                                docker-container:priceless_nash 0.0s
=> => reading samples/bookinfo/src/docker-bake.hcl 2.49kB / 2.49kB      0.0s
=> [internal] booting buildkit                                           28.7s
=> => pulling image moby/buildkit:buildx-stable-1                      28.7s
```

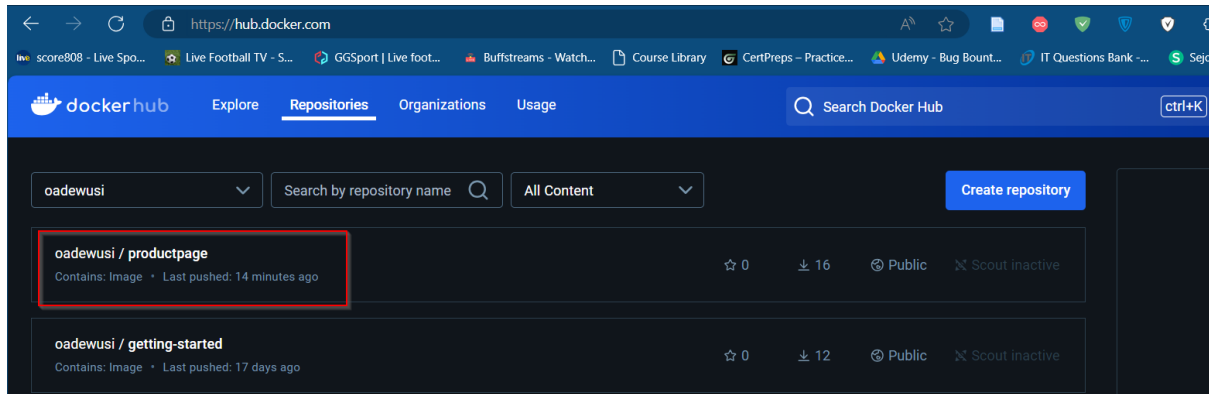
I used `docker build -t $BOOKINFO_HUB:$BOOKINFO_TAG .` to build the new image

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src/productpage$ docker build -t $BOOKINFO_HUB:$BOOKINFO_TAG .
[+] Building 60.4s (18/18) FINISHED
=> [internal] load build definition from Dockerfile
=> => pulling image docker:default
```

Inspecting the image built with `docker image ls` and pushing it to docker hub:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src/productpage$ docker image ls
REPOSITORY          TAG       IMAGE ID       CREATED        SIZE
oadewusi/productpage latest    d7f50732c9b3   2 hours ago   190MB
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src/productpage$ docker login
Authenticating with existing credentials...
Login Succeeded
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src/productpage$ docker push oadewusi/productpage:latest
```

Verification of the image successfully pushed to docker hub repo:



I edited the bookinfo.yaml to switch from the default image to my new image

```
! bookinfo.yaml X
istio-master > samples > bookinfo > platform > kube > ! bookinfo.yaml
306   spec:
312     template:
313       metadata:
318         labels:
320           version: v1
321     spec:
322       serviceAccountName: bookinfo-productpage
323       containers:
324       - name: productpage
325         image: docker.io/oadewusi/productpage:latest
326         imagePullPolicy: Always
327       ports:
328       - containerPort: 9080
329       volumeMounts:
330       - name: tmp
331         mountPath: /tmp
332       volumes:
333       - name: tmp
334         emptyDir: {}
335     ---
336
```

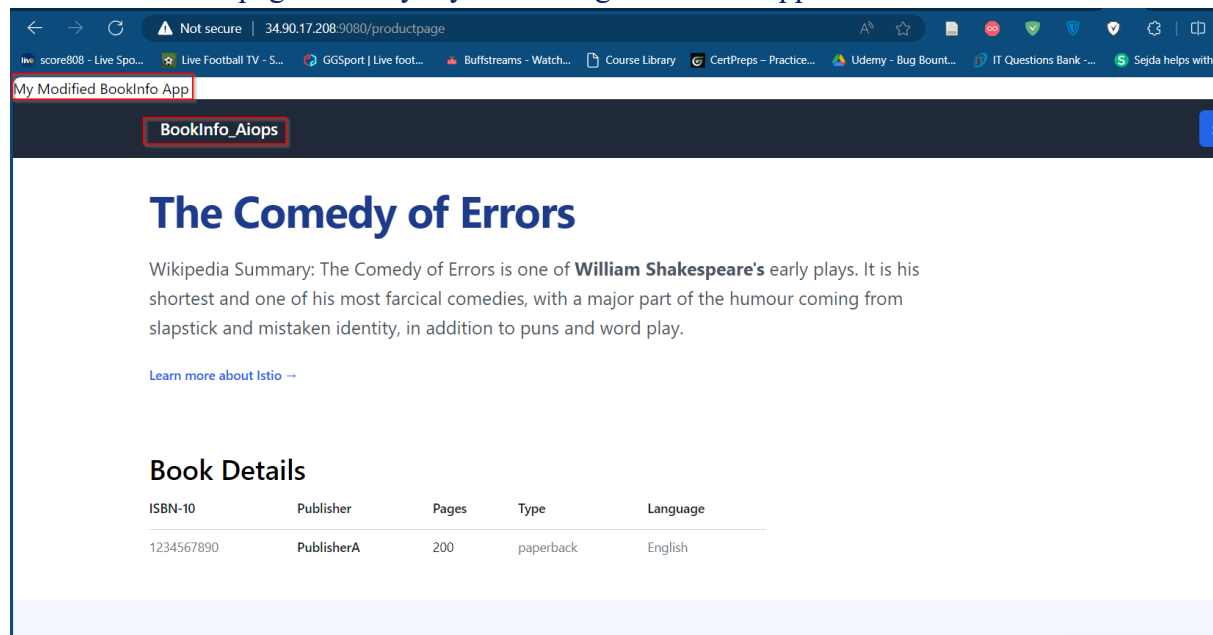
Redeploy my application by re-applying the Kubernetes.yaml file:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl apply -f platform/kube/bookinfo.yaml
service/details unchanged
serviceaccount/bookinfo-details unchanged
deployment.apps/details-v1 unchanged
service/ratings unchanged
serviceaccount/bookinfo-ratings unchanged
deployment.apps/ratings-v1 unchanged
service/reviews unchanged
serviceaccount/bookinfo-reviews unchanged
deployment.apps/reviews-v1 unchanged
deployment.apps/reviews-v2 unchanged
deployment.apps/reviews-v3 unchanged
service/productpage unchanged
serviceaccount/bookinfo-productpage unchanged
deployment.apps/productpage-v1 configured
```

Then issued a kubectl get pod command to view the updated productpage:

```
oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
details-v1-6cd6d9df6b-dt9tj        1/1     Running   0           11h
productpage-v1-6bf4f68bd6-cjbzg    1/1     Running   0           62s
ratings-v1-794744f5fd-fhmqb        1/1     Running   0           11h
reviews-v1-67896867f4-j6clg        1/1     Running   0           11h
reviews-v2-86d5db4bd6-2rdsx        1/1     Running   0           11h
reviews-v3-77947c4c78-mhmqh9       1/1     Running   0           11h
```

Reloaded the webpage to verify my new changes has been applied:



Then issued a describe pod command to check the new image attached:

```
oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl describe pods productpage-v1-6bf4f68bd6-cjbzg
Name:                productpage-v1-6bf4f68bd6-cjbzg
Namespace:           default
Priority:             0
Service Account:     bookinfo-productpage
Node:                gke-onlineboutique-default-pool-35c278d7-2wf5/10.164.0.3
Start Time:          Thu, 03 Oct 2024 00:03:03 +0200
Labels:              app=productpage
                    pod-template-hash=6bf4f68bd6
                    version=v1
Annotations:         prometheus.io/path: /metrics
                    prometheus.io/port: 9080
                    prometheus.io/scrape: true
Status:              Running
IP:                  10.76.0.20
IPs:                 10.76.0.20
Controlled By:       ReplicaSet/productpage-v1-6bf4f68bd6
Containers:
  productpage:
    Container ID:   containerd://3a9e1a3abf38450e1c92597feb16f696043d225c9a6a4624dc48d01cd057951f
    Image:          docker.io/oadevusi/productpage:latest
    Image ID:       docker.io/oadevusi/productpage@sha256:95093c55af05537f214d5d6fea716e7ec2bd61959ab68801bcb54b1e6d21f2bc
    Port:           9080/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Thu, 03 Oct 2024 00:03:09 +0200
    Ready:          True
    Restart Count:  0
    Environment:    <none>
```

Lab Task 3 – Monitor GKE with Prometheus/Grafana

Added Prometheus monitoring and a Grafana panel to track requests to the bookinfo ProductPage and edited the Prometheus config file in the sandbox “containers” directory for Prometheus and to add an entry for Bookinfo:

```
! config.yml X productpage.py
istio-master > samples > bookinfo > containers > prometheus > ! config.yml
5   scrape_configs:
18   - job_name: "app_one"
19     static_configs:
20       - targets:
21         - "app_one:8000"
22   - job_name: "monitor_app"
23     static_configs:
24       - targets:
25         - "monitor_app:8003"
26   - job_name: "bookinfo"
27     static_configs:
28       - targets: ['34.90.17.208:9080']
29
```

Examined the Python source code to the BookInfo product page to verify the metric:

```
! config.yml X productpage.py X
istio-master > samples > bookinfo > src > productpage > productpage.py
85
86   service_dict = {
87     "productpage": productpage,
88     "details": details,
89     "reviews": reviews,
90   }
91
92   request_result_counter = Counter('request_result', 'Results of requests', ['destination_app', 'response_code'])
93
```

Bringing up my compose file:

```
oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ docker compose up -d
[+] Running 26/20
✓ postgres_exporter Pulled
✓ node_exporter Pulled
✓ push_gateway Pulled
✓ postgres Pulled
```

Verification of the containers created:

```
✓ Network bookinfo_default Created
✓ Container prometheus_node_exporter Started
✓ Container prometheus_push_gateway Started
✓ Container grafana Started
✓ Container prometheus Started
✓ Container postgres Started
✓ Container postgres_exporter Started
oadevusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$
```

Navigated back to Grafana and to add the visualization for the product page metric. Hitting the product page a number of times to generate some manual load.

request_result_created metric and request_result_total metic for 5 mins



request_result_created metric and request_result_total metic for 10 mins



request_result_created metric and request_result_total metic for 15 mins

